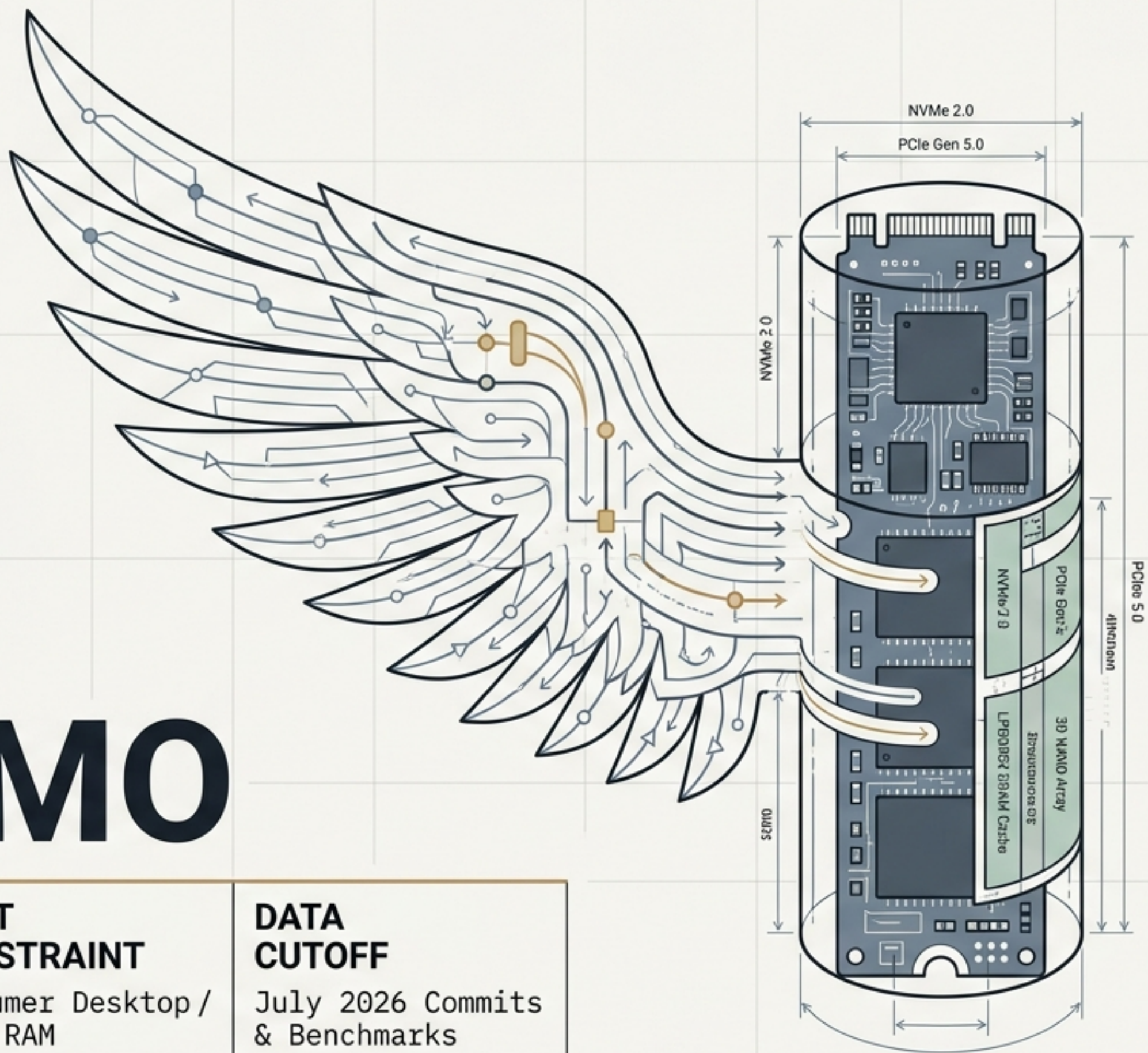


PENG-MIMO



CORE ENGINE

Colibri-style streaming MoE

TARGET ARCHITECTURE

MiMo-V2.5 / 311B Parameters

HOST CONSTRAINT

Consumer Desktop / 32GB RAM

DATA CUTOFF

July 2026 Commits & Benchmarks

Riding the Digital Whirlwind

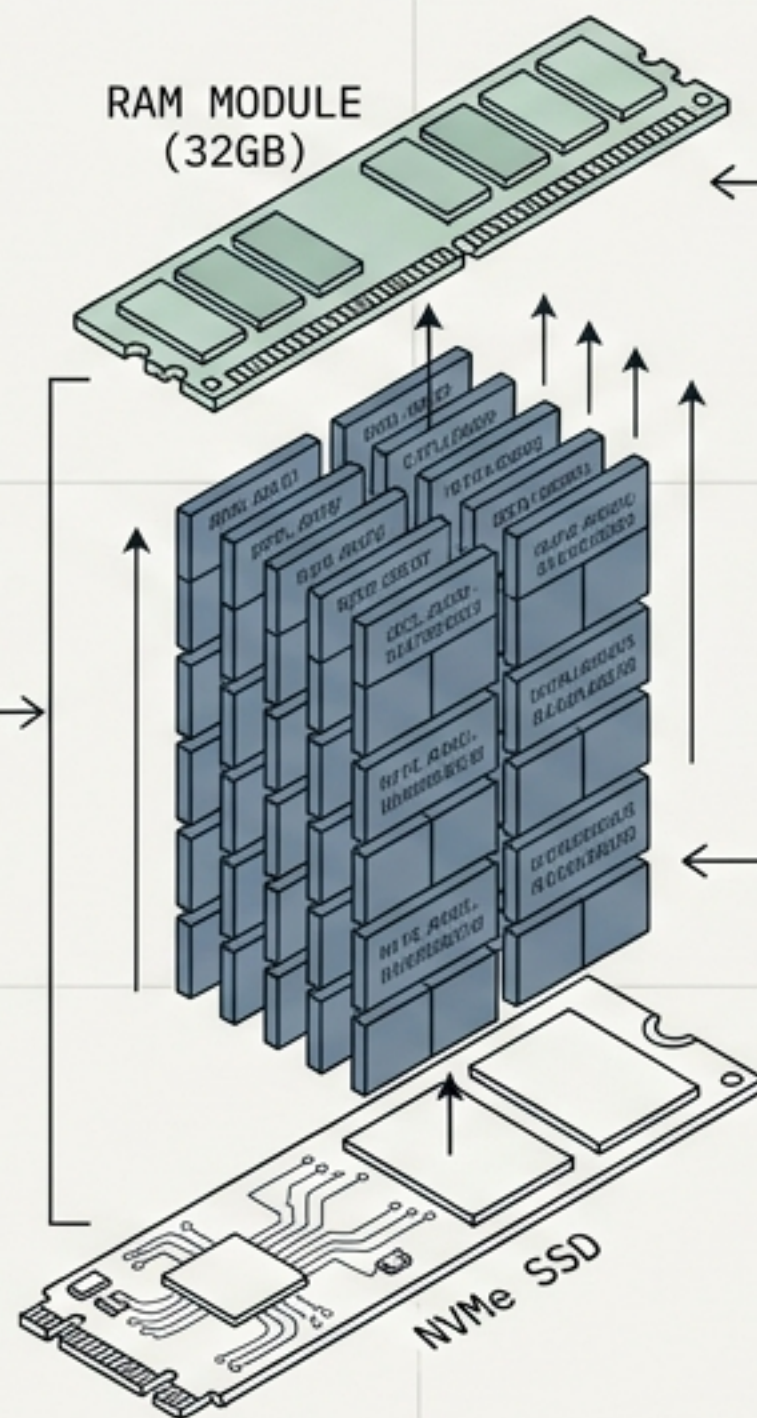
The Myth



In the *Zhuangzi*, the mythological Peng bird is a creature of immense size. It cannot fly through normal aerodynamics; it stays airborne solely by riding a colossal whirlwind.

The Engineering Reality

The Bird:
A 311 Billion parameter language model.



The Constraint:
A consumer desktop with only 32GB of RAM.

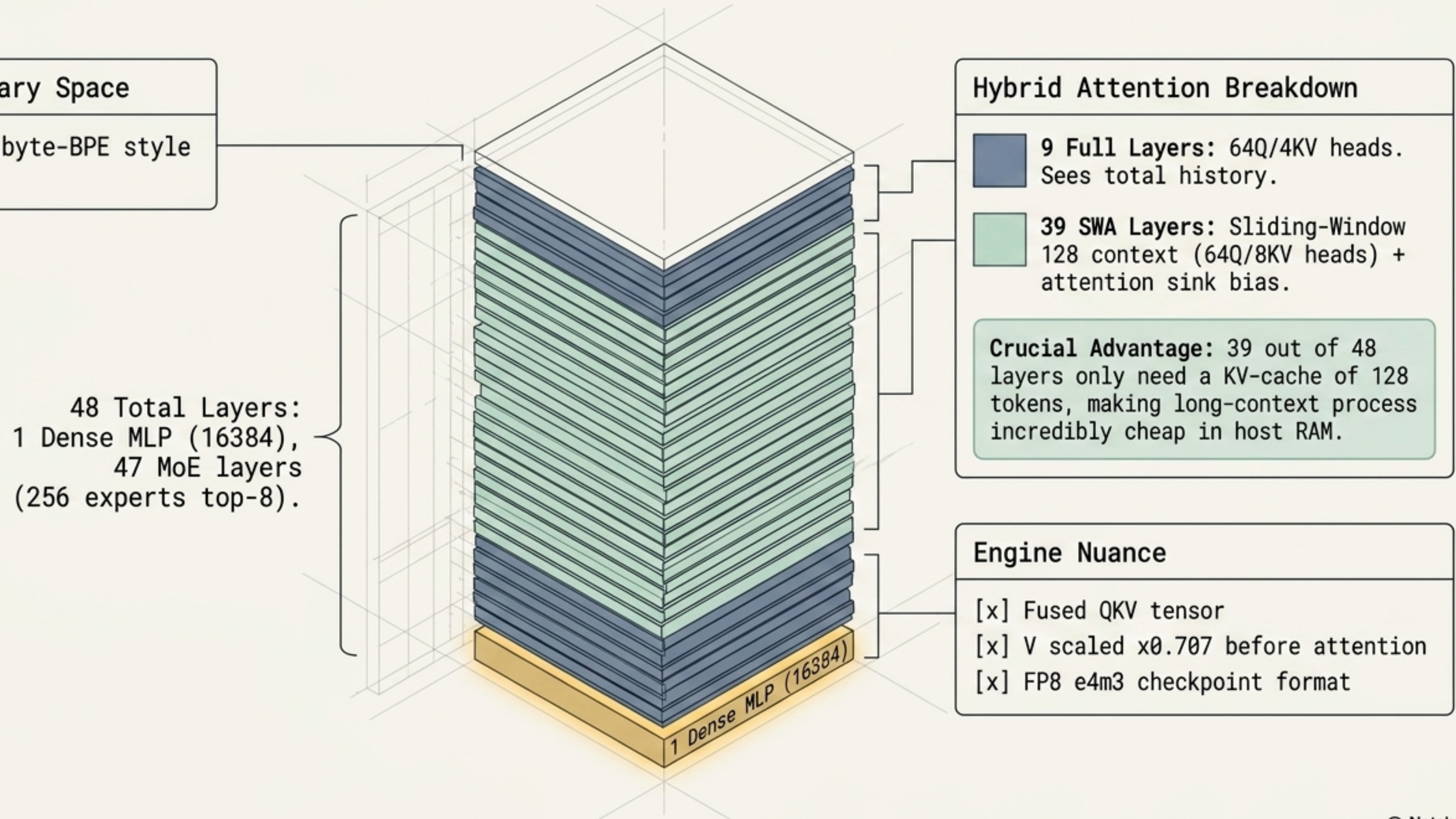
The Whirlwind:
Peng-MIMO streams ~4.7 GB of localized NVMe reads per cold token to keep the model operational, pulling only the necessary routing experts into memory on demand.

Maximizing Colibri Architecture Reuse While Respecting Physical Constraints

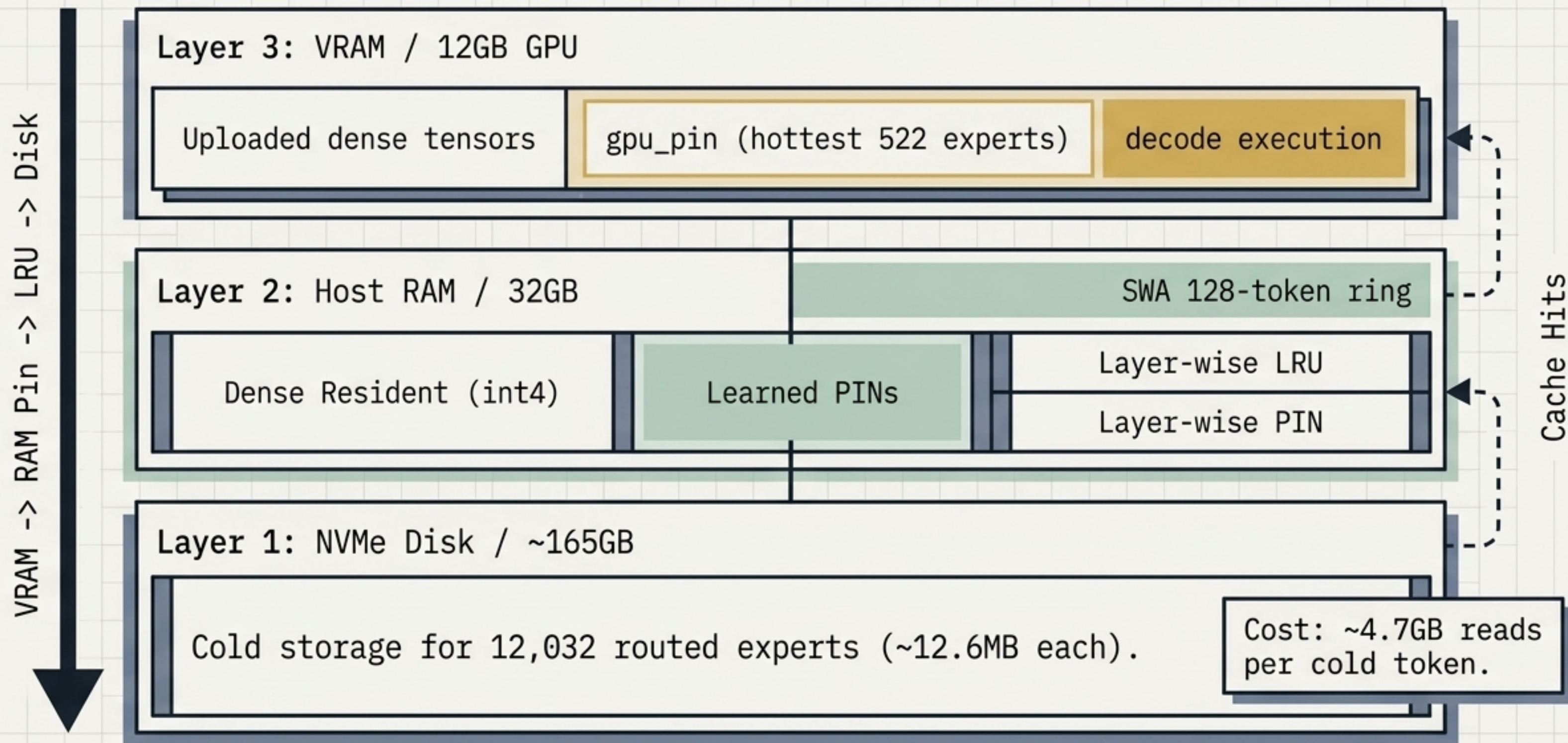
Metric	MiMo-V2.5	GLM-5.2	Hy3	Qwen3.5-122B
Disk Size (Int4)	~165 GB (Fits)	~370 GB (Fails)	~157 GB	~65 GB
Active Params / Token	15B	40B	21B	10B
Router Type	Identical to Colibri	Identical	Different	Different
Attention Mechanism	Simple GQA	MLA+DSA	GQA (Hetero)	Gated DeltaNet
Checkpoint Format	FP8 128x128	-	Different	Different

Perfect footprint. At 165GB total and only 15B active parameters, it fits the consumer NVMe ceiling perfectly without requiring a complex C-based router rewrite.

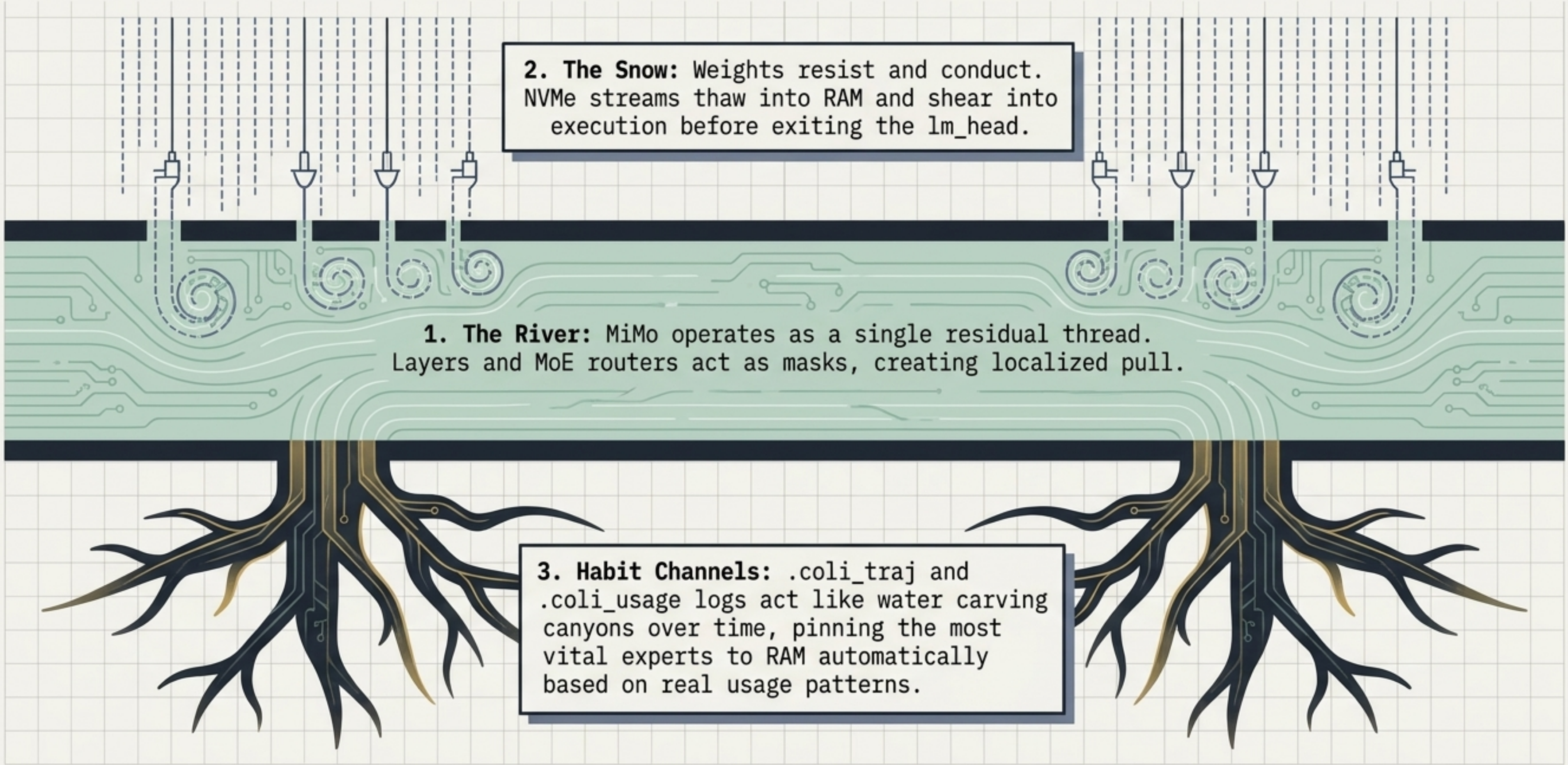
The Logical Geometry of MiMo-V2.5



The Physical Lookup Priority



'Corriente Peng': The Residual River



The diagram illustrates the 'Corriente Peng' architecture, which is visualized as a river system. At the top, vertical dashed lines represent data streams entering the system. These streams pass through a layer of 'snow' (weights) and then enter a central 'river' (residual threads). The river flows horizontally across the middle of the diagram, with various circuit-like patterns and swirls indicating localized pull and execution. Below the river, a network of 'habit channels' (usage logs) is shown, which act like water carving canyons over time. The diagram is set against a light gray grid background.

2. The Snow: Weights resist and conduct. NVMe streams thaw into RAM and shear into execution before exiting the `lm_head`.

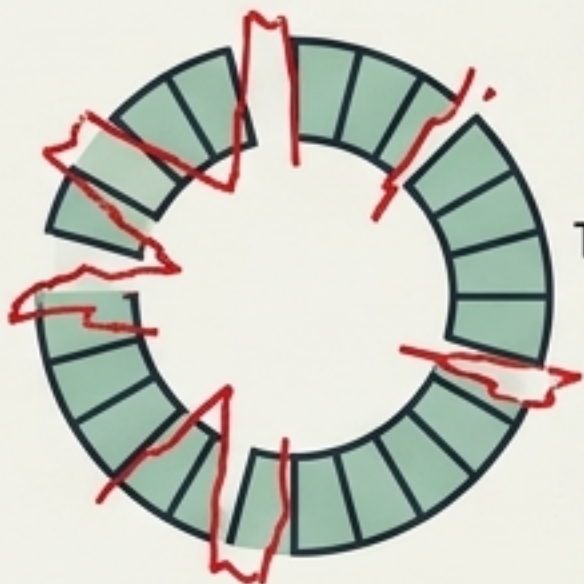
1. The River: MiMo operates as a single residual thread. Layers and MoE routers act as masks, creating localized pull.

3. Habit Channels: `.coli_traj` and `.coli_usage` logs act like water carving canyons over time, pinning the most vital experts to RAM automatically based on real usage patterns.

July 2026 Core Commits & Optimizations

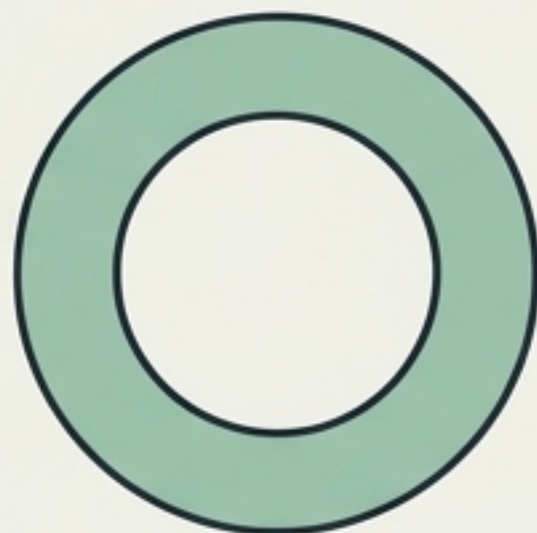
mimo: fix SWA ring batch-prefill corruption

Corrupted Ring Buffer
(SWA Batch-Prefill)



Transformation

Resolved
Ring Buffer

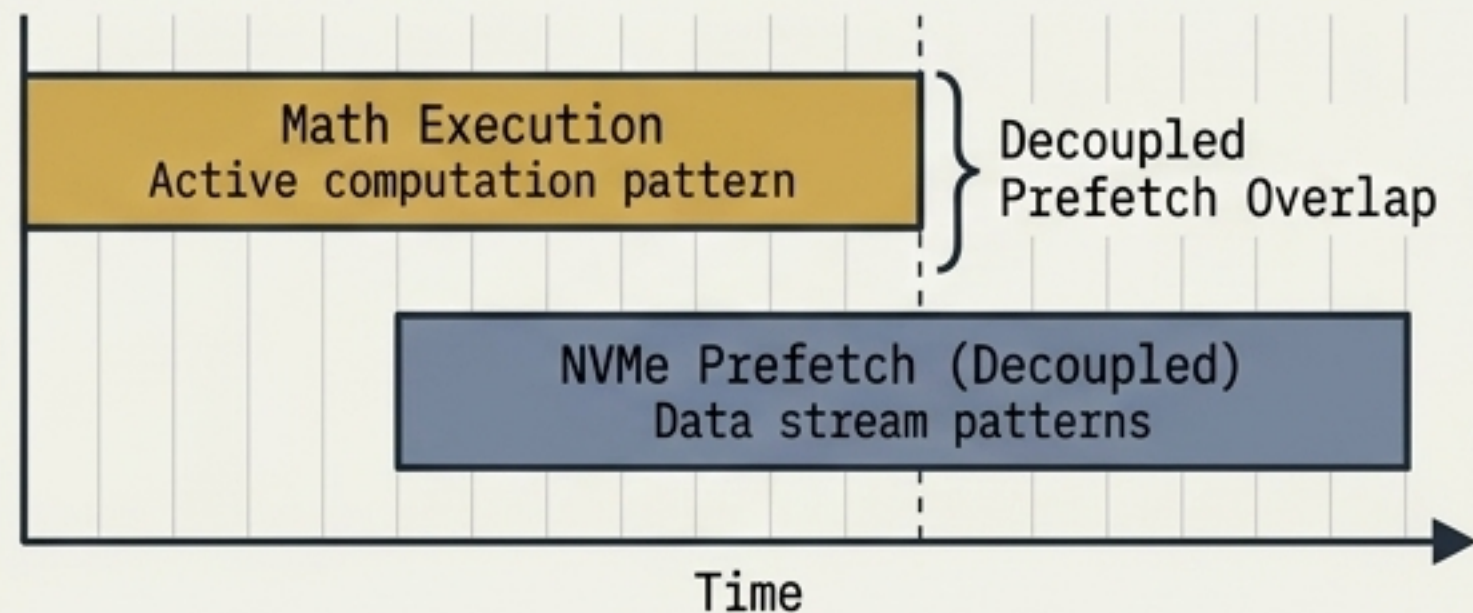


Corrupted Ring Buffer
(SWA Batch-Prefill)

Resolved
Ring Buffer

Impact: Resolved memory corruption in Sliding Window Attention ring buffers during the critical batch prefill phases.

mimo: PILOT/LOOKA prefetch + RoPE table



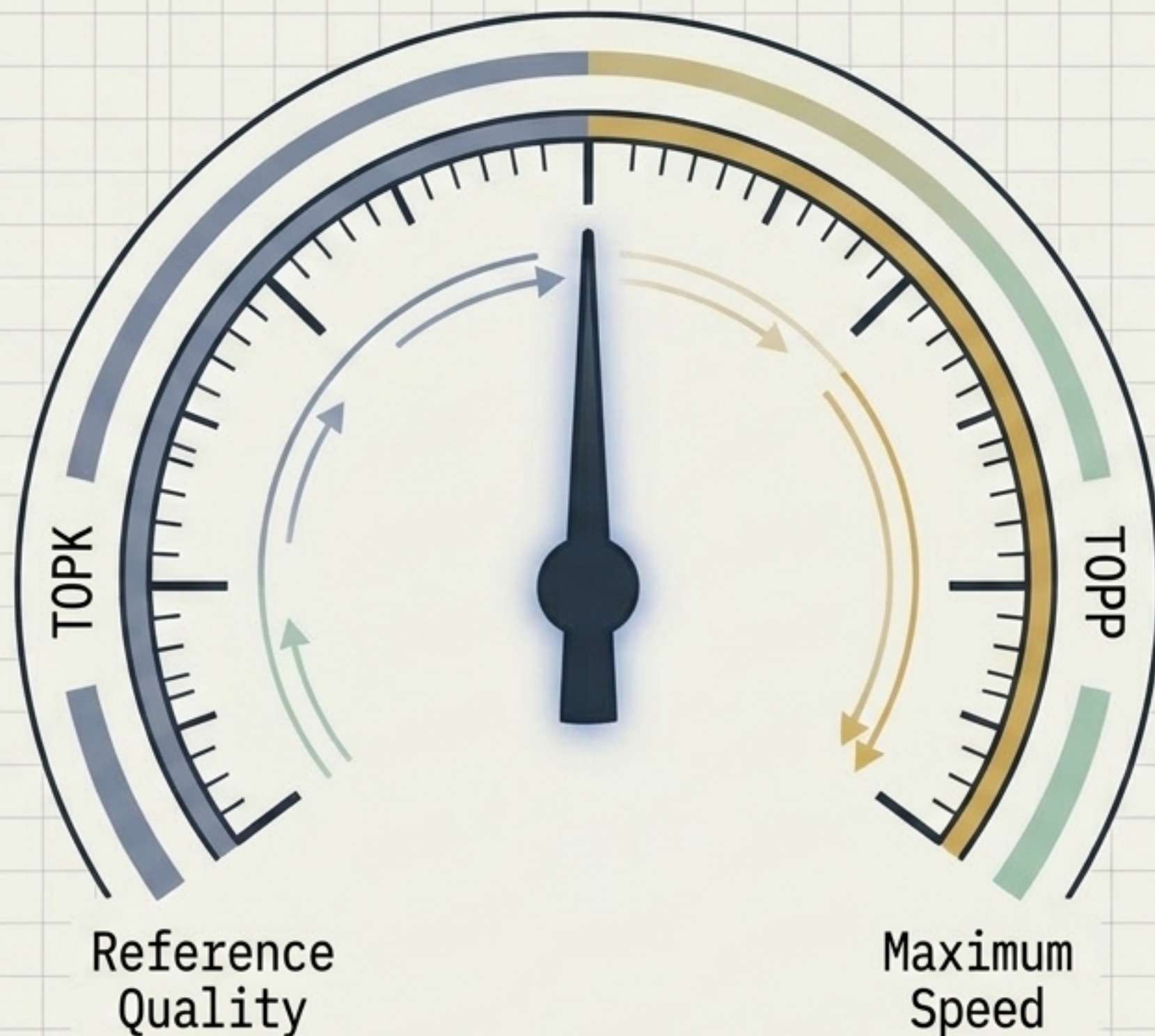
Mechanism: Overlaps disk I/O by decoupling expert prefetching from math execution.

The Result: At RAM-starved limits (cap->3), PILOT yields a massive **~1.3x to 2.0x speedup** purely by hiding NVMe latency in the background.

The Rule of the Frontier: Nothing is Free

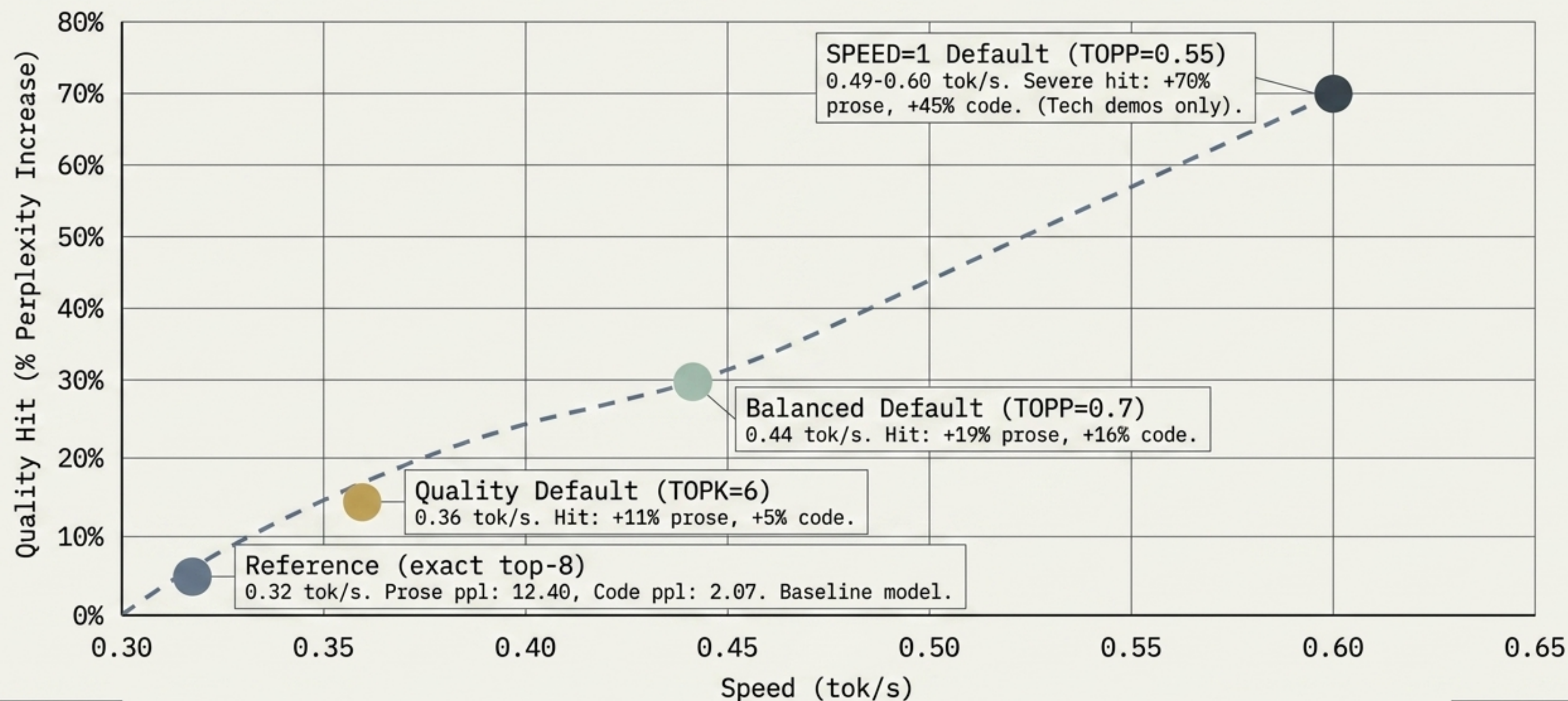
To push beyond hardware physics, Peng-MIMO utilizes expert trimming knobs.

The Iron Law:
Generation speed tracks the volume of experts loaded almost linearly.



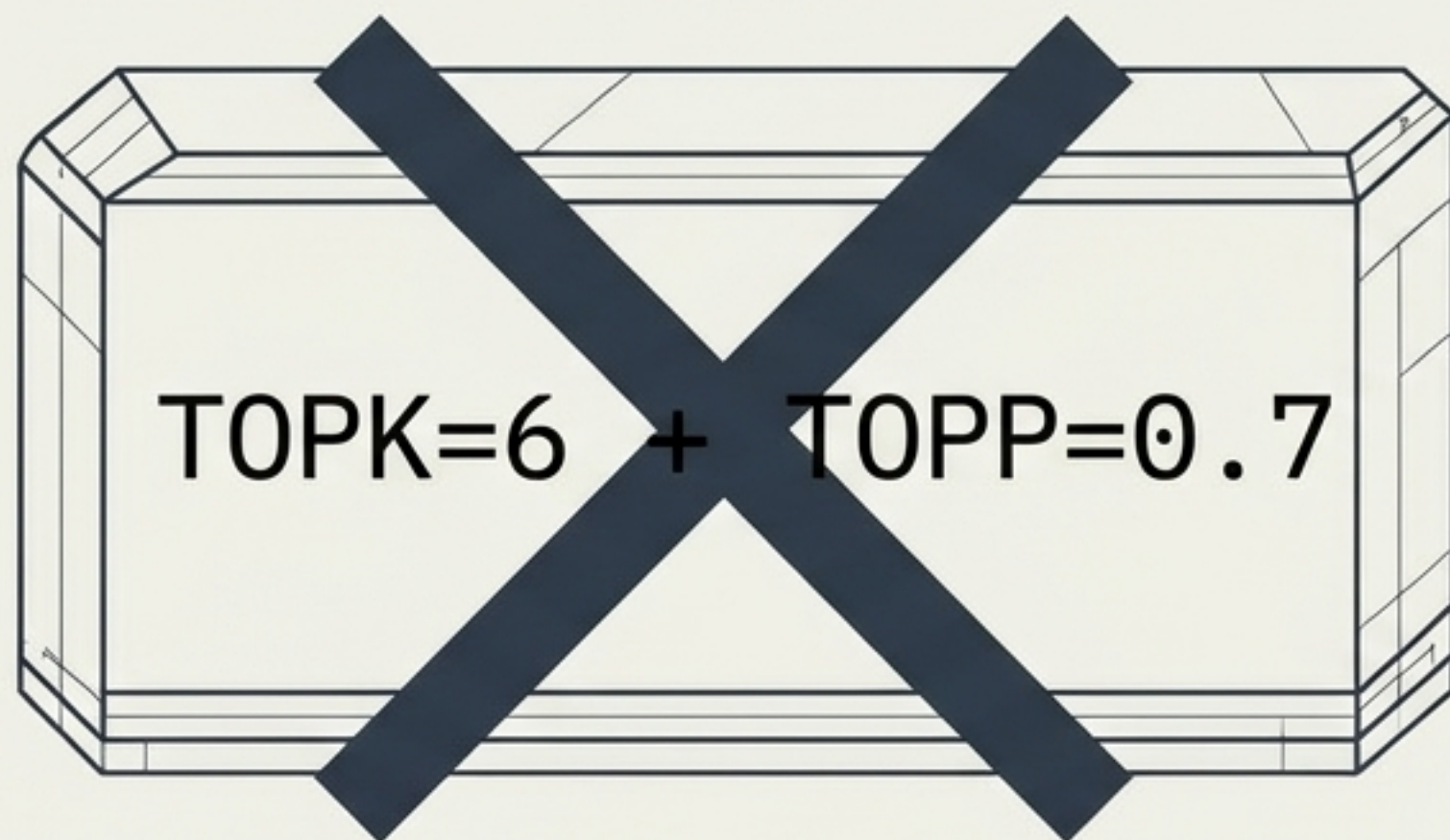
Methodology Note:
All measurements on the frontier (July 13th) were evaluated via teacher-forcing perplexity on a fixed 1,534-token prose/code corpus to objectively measure the exact cost of speed.

Mapping the Quality vs. Speed Frontier



The Discarded Hybrid Approach

The Experiment

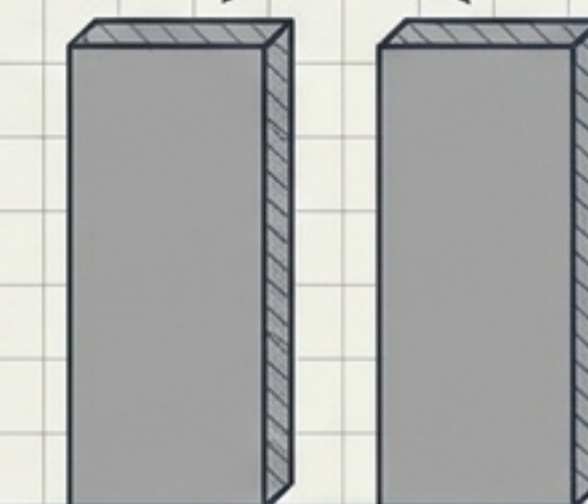


Combining top-K trimming with cumulative probability (top-P) trimming to aggressively cull experts (July findings 40.3). Formally discarded from the codebase.

The Post-Mortem

Speed Gain

Zero speed gain.
Redundant logic
yielded no additional
NVMe I/O savings.

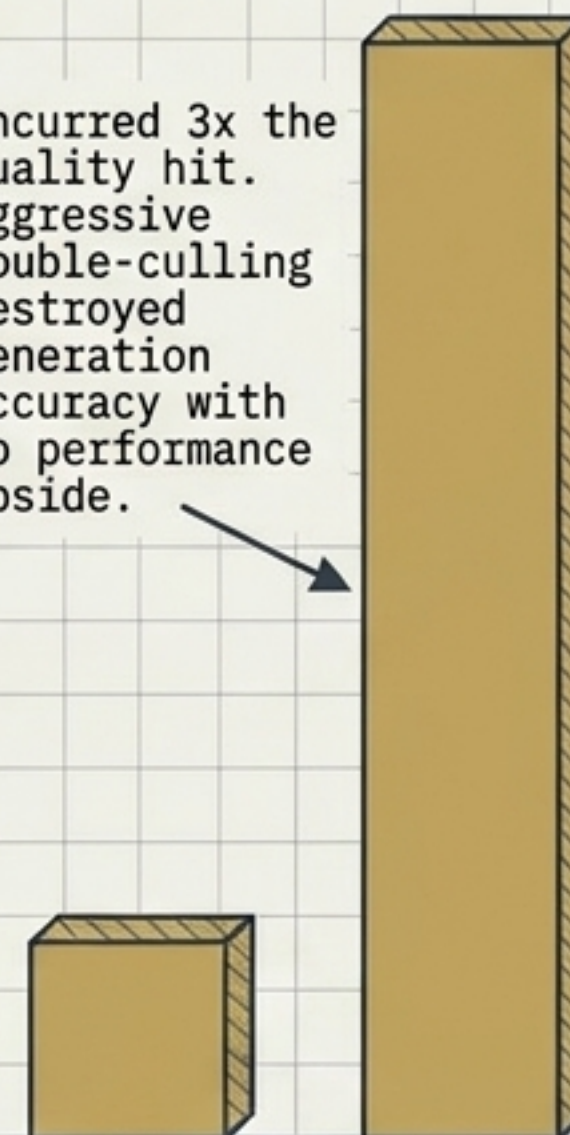


Standalone
TOPP=0.7

Hybrid

Quality Penalty

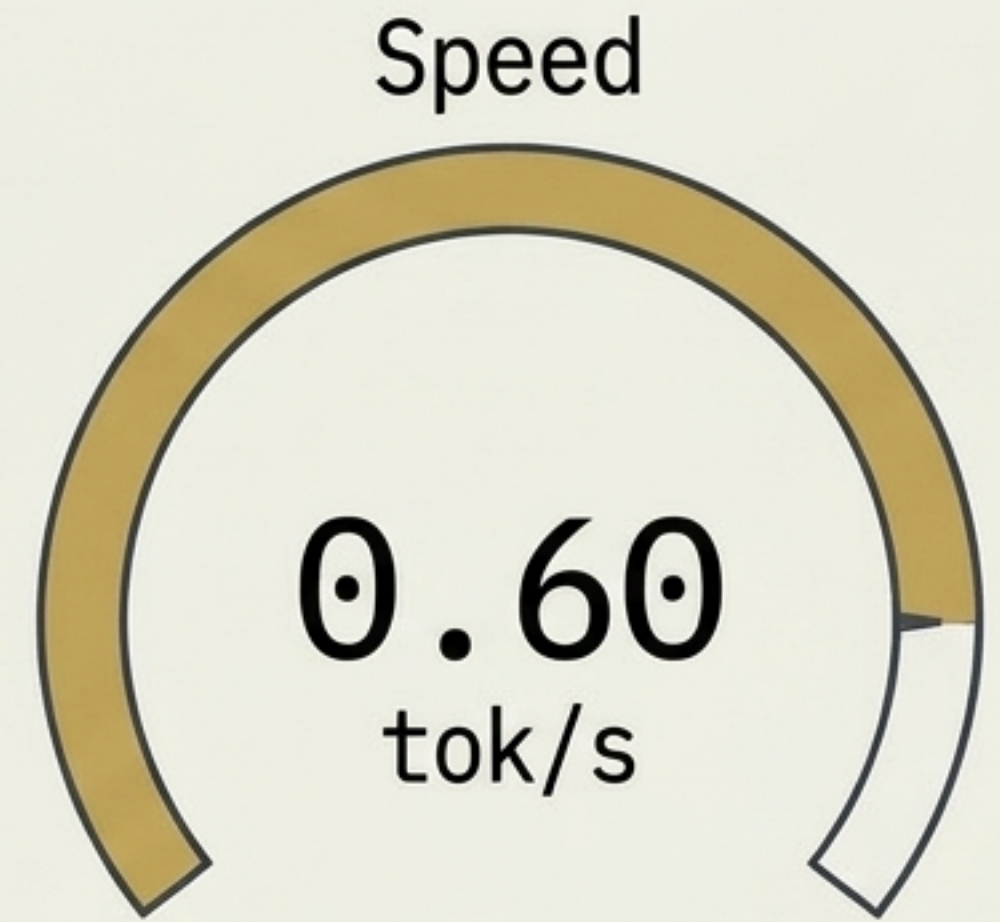
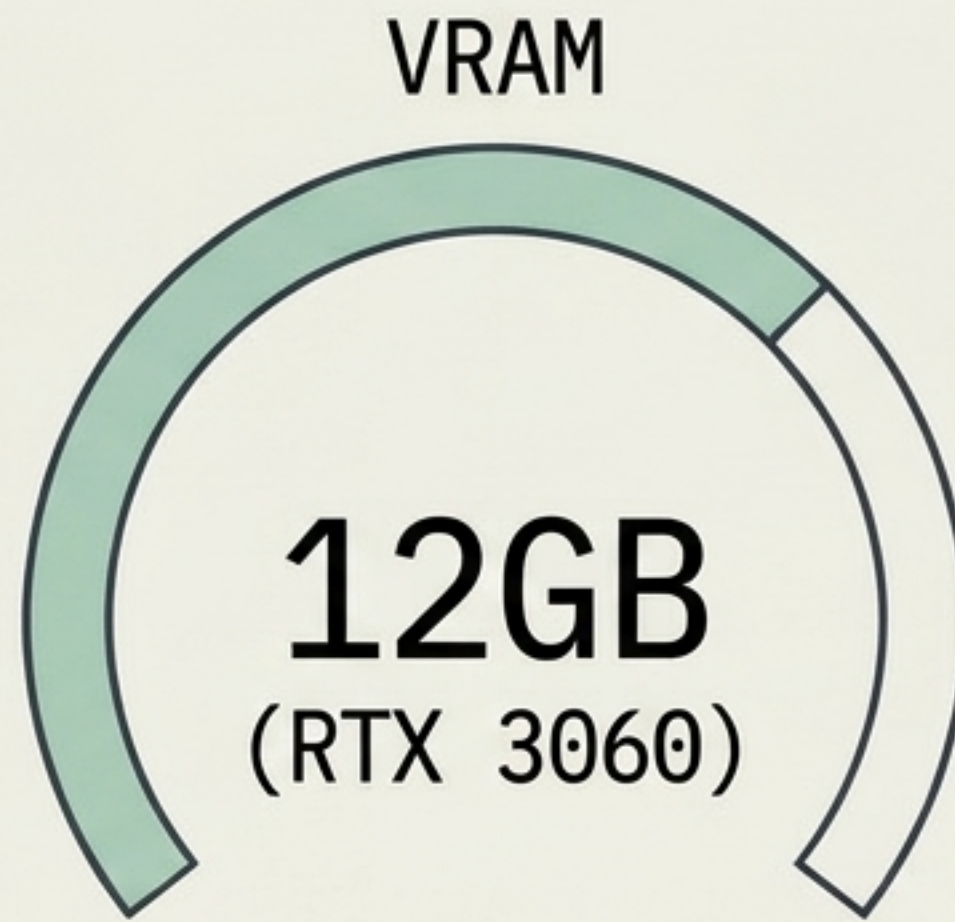
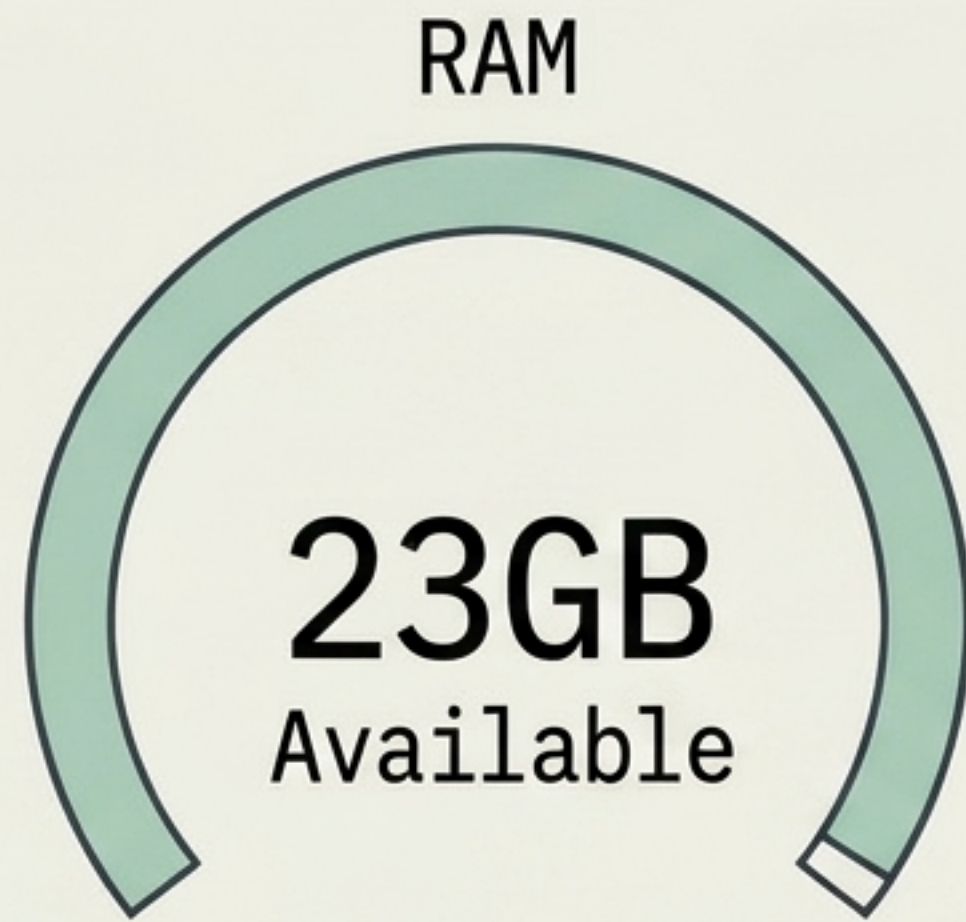
Incurring 3x the
quality hit.
Aggressive
double-culling
destroyed
generation
accuracy with
no performance
upside.



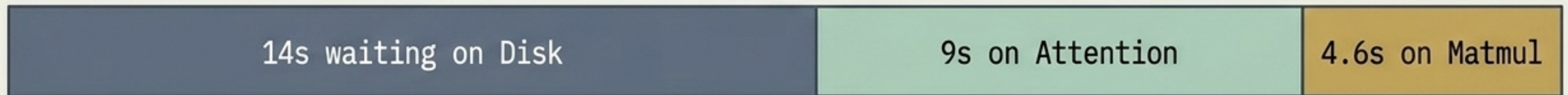
Standalone

Hybrid

Real-World Performance Limits (July 12 Record)



Execution Profile (cap=3, disk-bound)

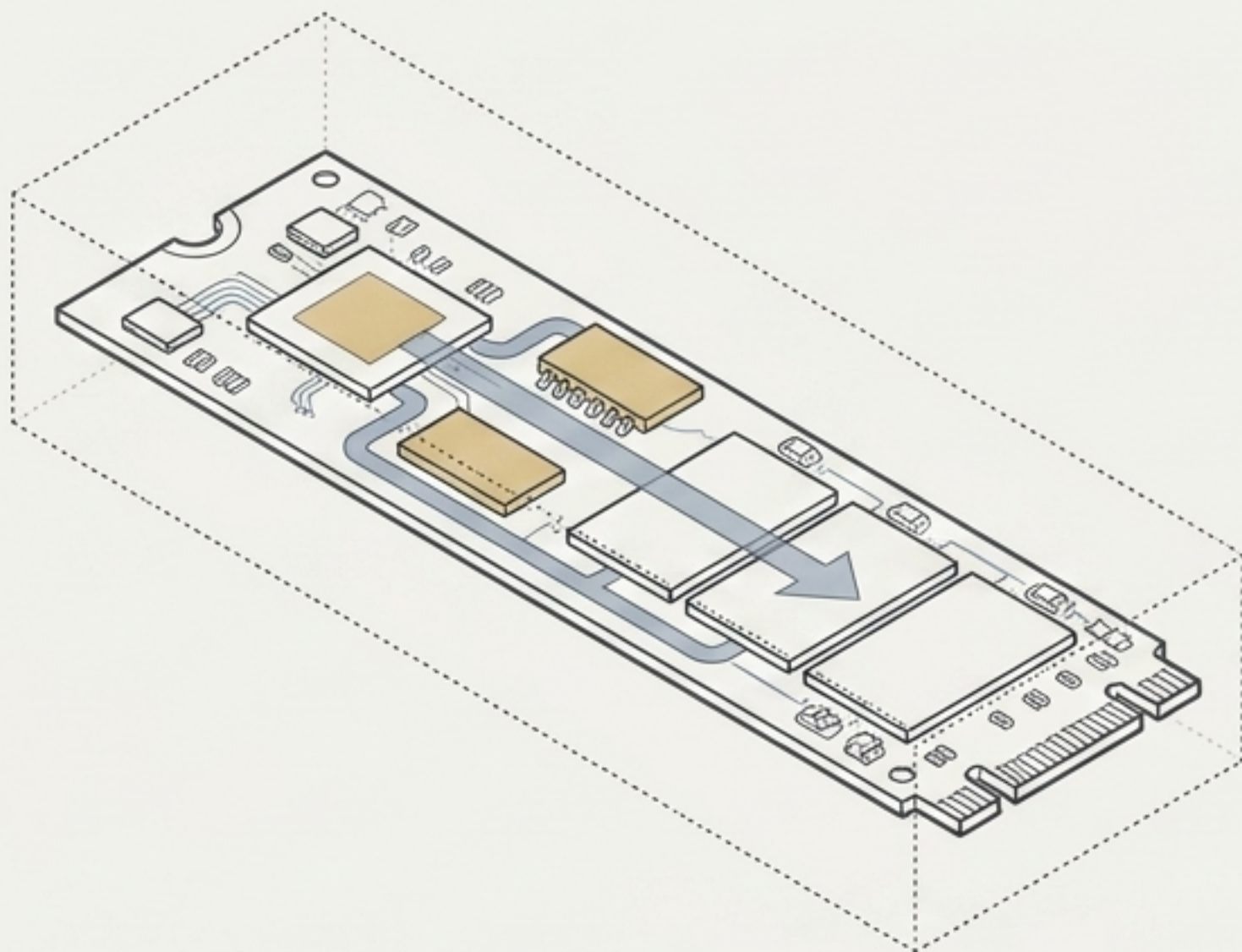


Intel Xeon W-2140B (8C/16T) | WSL2 NVMe (ext4)



At 23GB RAM, the engine drops to cap=3 (holds ~3 experts/layer). The operation becomes completely disk-bound.

The Physics of SSD Endurance



**Does pulling 4.7GB per token
destroy a consumer SSD?**

Peng-MIMO Load vs. Drive Wear Metric (TBW)

Incredibly Heavy
Read Stream
(~4.7GB/token)

Near-Zero Write
Wear (Negligible
log/.coli_usage)



The Engineering Reality:

Consumer NVMe endurance is strictly rated in Terabytes Written (TBW).

Peng-MIMO generates an incredibly heavy read stream, which does not degrade TBW.

Thermal Warning:

While drive wear is safe, sustained multi-TB reads over long sessions may induce thermal throttling.

The Path to the 1.0 tok/s Barrier

Lever 1: Hardware Scaling

More RAM. The ultimate physical fix. Scaling to 32-64GB RAM allows the LRU cache to expand, eventually eventually reducing disk stalls to zero.

**1.0
tok/s**

Lever 3: Algorithmic

Native MTP. Integrated but currently off by default. Speculative drafting requires excess RAM for a warm expert cache to offset the heavy I/O cost of validation batches.

Lever 2: Environment Escapes

Native Linux. Removing the WSL2 VHDX virtualization overhead unleashes the raw, bare-metal O_DIRECT NVMe speeds.